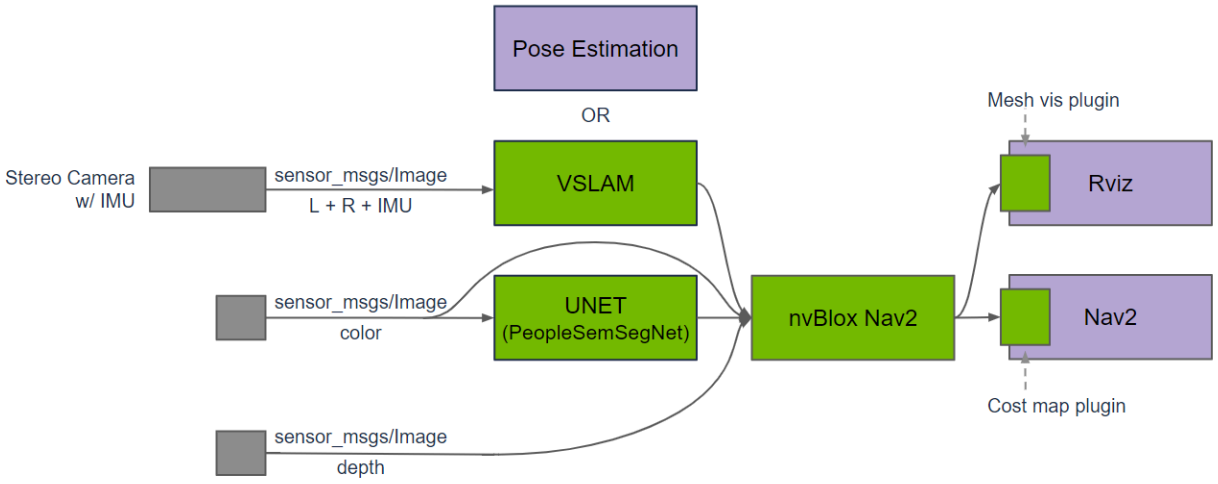
Isaac ROS 3D scene reconstruction and mapping

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- Overview
- Quick experience

Isaac ROS Nvblox official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_nvblox/index.html

Overview

Nvblox reconstructs 3D scenes in real time from depth and pose data, and can output a 2D cost map for Nav2. It works with a depth camera or a 3D LiDAR and provides an environmental representation for robot navigation.

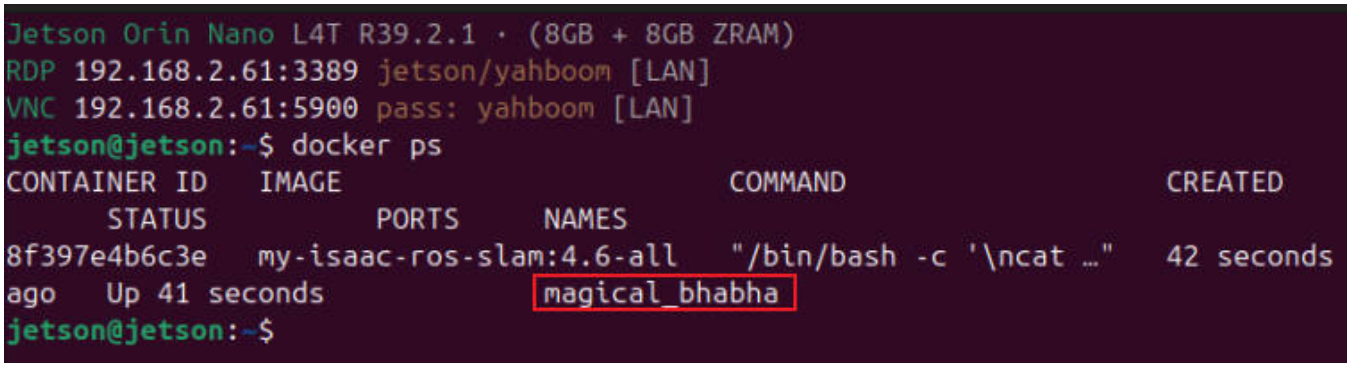


Quick experience

First start the container with the following command

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open the container terminals with `docker exec -it <container name> bash`. Each terminal has to use the same ROS/FastDDS environment, for example:



Note: all the terminals below are executed inside the container:

Terminal A: start Nvblox:

```
ros2 run nvblox_ros nvblox_node --ros-args \
  --params-file /opt/ros/jazzy/share/nvblox_examples_bringup/config/nvblox/nvblox_base.yaml
\
  --params-file
/opt/ros/jazzy/share/nvblox_examples_bringup/config/nvblox/specializations/nvblox_sim.yaml \
  -p use_sim_time:=false \
  -p global_frame:=odom \
  -p map_clearing_radius_m:=-1.0 \
  -p layer_visualization_exclusion_radius_m:=0.0 \
  -p decay_tsdf_rate_hz:=0.0 \
  -p num_cameras:=1 \
  -r camera_0/depth/image:=/front_stereo_camera/depth/ground_truth \
  -r camera_0/depth/camera_info:=/front_stereo_camera/left/camera_info \
  -r camera_0/color/image:=/front_stereo_camera/left/image_raw \
  -r camera_0/color/camera_info:=/front_stereo_camera/left/camera_info
```

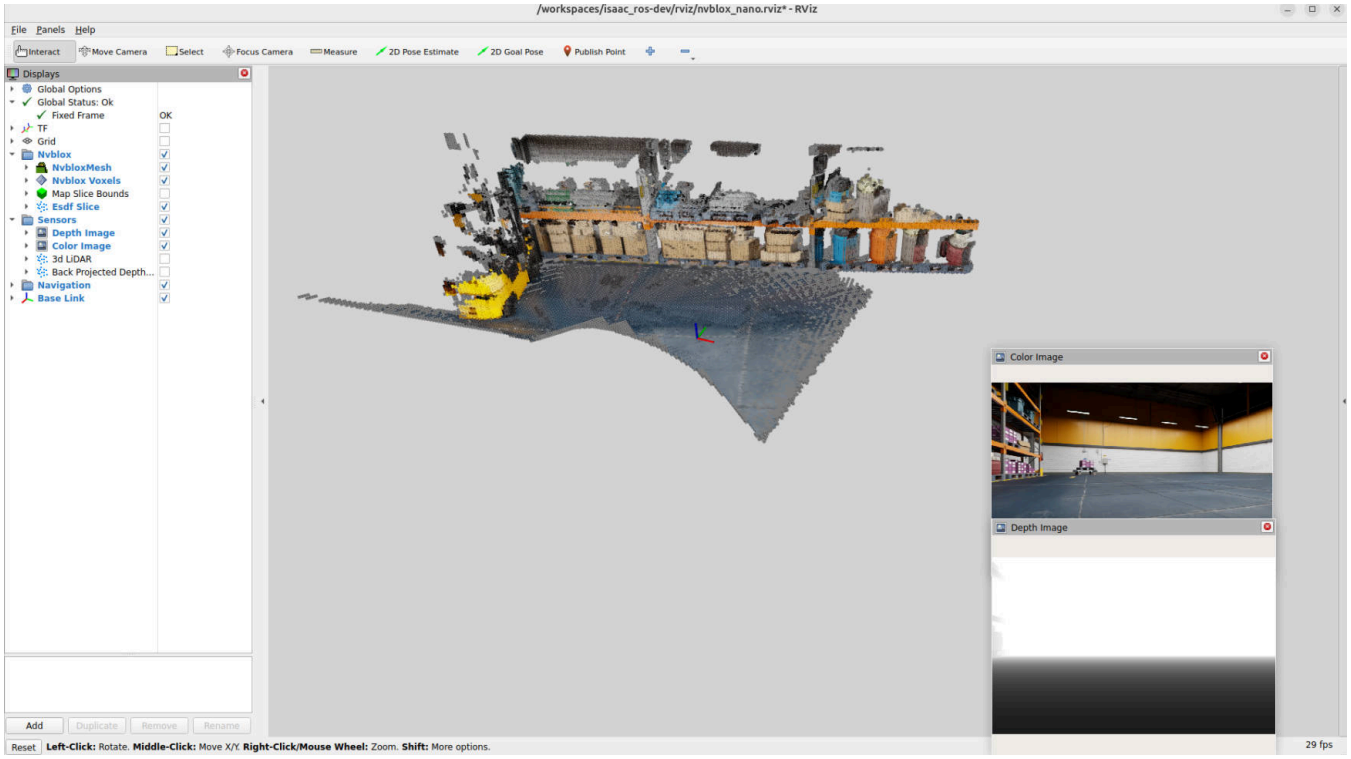
Terminal B: open the dedicated RViz configuration:

```
rviz2 -d /workspaces/isaac_ros-dev/rviz/nvblox_nano.rviz
```

The configuration displays `/nvblox_node/mesh`, `/nvblox_node/color_layer` and the ESDF.

Terminal C: play the quickstart bag only once:

```
ros2 bag play /workspaces/isaac_ros-dev/isaac_ros_assets/isaac_ros_nvblox/quickstart
```



After the playback finishes, the bag exits on its own; Nvblox and RViz keep running, the image stays on the last frame, and the map in memory continues to be displayed. You can open another terminal to check:

```
ros2 topic echo --once /nvblox_node/mesh
ros2 topic echo --once /nvblox_node/static_map_slice
```